Quantum Variational Eigensolvers on Adiabatic Quantum Computing Devices

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1 Motivation

Since the development of the quantum computers, many claims regarding quantum advantage have been made. Often, these advantage claims did not last long. In this work, we focus exclusively on the D-Wave Quantum Annealer, which leverages quantum technology to solve quadratically unconstrained binary optimization (QUBO) problems. D-Wave is among the first companies to commercially sell quantum computers. In 2025, researcher from D-Wave claimed to have solved a scientifically relevant problem fast using quantum technology, than it would have been possible on classical devices. However, it is currently an open question whether the D-Wave Quantum Annealer is able to solve other problems well, and possible faster than convential computers. In this work, we assess the D-Wave's ability to sample from a probability distribution, to approximate the ground state of different statistical mechanical models using variational algorithms.

1.1 Previous Work

The D-Wave's Quantum Annealers performance scaling for finding the ground state (corresponding to the minimum of the quadratic binary cost function) has been extensively studied, with classical methods exhibiting better scaling and solution times. This has been done on a wide range of problems, ranging from cryptographic instances to physics inspired models.

2 Methods

In this section we briefly outline the main computational and algorithmic results used to gather results.

2.1 Sampling Methods

We use classical and quantum state-of-the-art methods to sample from the target distribution.

Quantum Annealing Quantum Annealing is inspired by

3 Variational Algorithms

Variational algorithms are a class of hybrid quantum-classical optimization methods. We focus on the Variational Quantum Eigensolver (VQE) which is designed to find the ground state of a given Hamiltonian. It consists of three parts: the *cost function* to be minimized, the *ansatz*, which in our case is a Boltzman machine, and an *expectation value estimation*. Additionally, a classical optimizer is needed to improve the ansatz at each iteration. Depending on the quantum architecture, two approaches for estimating the expectation value are possible:

- On gate-based quantum computers, measuring can be used to gather expectation values of the modified ansatz and improve it further
- In adiabatic quantum computing, the quantum device is used to generate representative samples of the modified ansatz. These samples are then used to approximate the expectation value.

Mathematically, the problem can be described as follows: The goal is to approximate the ground state, that is, the eigenstate corresponding to the lowest eigenvalue of a Hamiltonian H. In physics, a Hamiltonian is defined to be an operator that describes the energy of a system. Efficient methods exist to encode many classical problems, such as scheduling or integer factoring, into Hamiltonians. The ground state of this Hamiltonian is often used to represent the problem's solution. Calculating this solution analytically using for example exact diagonalisation is computationally difficult, as the Hilbert space grows exponentially with the size of the quantum state. In both adiabatic and gate-based models, the ansatz can be represented as a unitary $U_{var}(\sigma)$, where σ describes some parametrization of this unitary. In the case where the quantum computer is initialized in the